

BWRR 1013

RISK AND INSURANCE

CHAPTER 1: INTRODUCTION TO RISK



Risk and
chance

Hazards
and perils

Objective
risk and
subjective
risk

Classifica-
-tion of
risk

Burden of
risk on
Society

1.1 RISK AND CHANCE



The chance of loss

The possibility of loss

Uncertainty

The dispersion of actual from expected results

The probability of any outcome different from the one expected



Risk is an **uncertainty concerning the occurrence of a loss**

i.e. the multiplicity of possible outcomes.

If a loss is certain to occur, it may be planned for in advance and treated as a definite, known expense.

When there is uncertainty about the occurrence of a loss, risk becomes an important problem.

Risk of being killed in an auto accident

Risk of lung cancer for smokers

Risk of being failed in the examination

1.2 RISK COMPONENTS



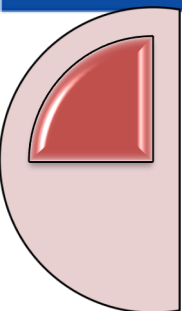


Peril

A potential cause of a loss, e.g., fires, hurricanes, theft, heart attack, and etc.

Hazard

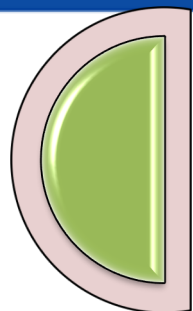
A condition that increase the probability of a loss, e.g., potential healthcare losses.



Physical hazard

e.g. A wooden building vs. a concrete building.

e.g. The presence of gas tank in a room



Moral hazard

e.g. Faking accidents, inflating claim amounts

e.g. Murdering the insured to collect the life insurance proceeds

e.g. Concealing health information during application for insurance

An insurance company may be exposed to moral hazard when its prospective policyholder did not disclose material facts



Attitudinal Hazard (Morale Hazard)

- **Carelessness or indifference to a loss**, which increases the frequency or severity of a loss.
- e.g. Leaving car keys in an unlocked car, which increases the chance of theft
- e.g. A careless driver vs. careful driver). A careless driver is more likely to get involved in accident.

Legal Hazard

- Refers to **characteristics of the legal system or regulatory environment** that increase the frequency or severity of losses
- e.g. discount for traffic summons vs. heavier compound for traffic summons.
- The discount given to the traffic offenders in order to induce them to forward to pay is legal hazard.

Relationship between Hazard, Peril and Loss



PERIL	HAZARD	LOSS
Lung cancer	Excessive smoking	* Medical expenses * Loss of income
Fire /explosion	Explosive material left unattended	* Property damage * Loss of income
Car accident	Poor brake	* Property damage * Loss of use * Loss of income
Negligence	Carelessness	* Court awards * Legal expenses

1.3 OBJECTIVE AND SUBJECTIVE RISK (CHANCE OF LOSS)



Chance of Loss: Probability that an event will occur

Objective probability

Examine past data and use mathematical equations to determine the likelihood of an event

Can be determined by deductive or inductive reasoning

Subjective probability

An individual's personal estimate of the chance of loss.